

Direct Head-to-Head: RLSTCcode vs Q-RLSTC

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All Models Ranked by Competitive Ratio

Rank	Model	System	Params	Val CR	Optimizer	Training Data
1	RLSTCcode (best, 3k)	RLSTCcode	514	0.5892	SGD	3,000 trajts
2	RLSTCcode (5-fold CV)	RLSTCcode	514	0.7543 ± 0.0369	SGD	4,000 trajts
3	VQ-DQN (5q $\times$ 3L)	Q-RLSTC	34	1.4811	SPSA	30 trajts
4	RLSTCcode (model-state4)	RLSTCcode	514	1.5453	SGD	unknown trajts
5	Control C (h=32 $\times$ 32)	Q-RLSTC	1314	1.6390	SPSA	30 trajts
6	Control B (h=64)	Q-RLSTC	514	5.1133	SPSA	30 trajts
7	Control A (linear)	Q-RLSTC	12	9.2051	SPSA	30 trajts

Key Comparisons

1. Same Architecture (Dense 5 $\rightarrow$ 64 $\rightarrow$ 2, 514 params)

Model	Val CR	Optimizer	Data	Difference
RLSTCcode	0.5892	SGD	3,000	—
Q-RLSTC Control B	5.1133	SPSA	30	8.7 $\times$ worse

The 514-param MLP works well with SGD+backprop (3k data) but fails under SPSA (30 data). The optimizer and data, not the architecture, determine performance.

2. Quantum vs Best Classical Control (same SPSA, same 30 trajs)

Model	Params	Val CR	CUT%	Advantage
VQ-DQN	34	1.4811	33.2%	best under SPSA
Control C	1314	1.6390	3.9%	39× more params

VQ-DQN achieves **9.6% lower CR** with **39× fewer parameters**. This demonstrates quantum parameter efficiency under gradient-free optimization.

3. Absolute Best Classical vs Quantum

Dimension	RLSTCcode (best)	VQ-DQN
Val CR	0.5892	1.4811
Parameters	514	34
Training data	3,000 trajs	30 trajs
Optimizer	SGD (gradient)	SPSA (gradient-free)
CR ratio	1.0×	2.5×
Data ratio	1.0×	100× less
Param ratio	1.0×	15× fewer

RLSTCcode wins on absolute CR (0.5892 vs 1.4811) thanks to 100× more training data and gradient-based optimization. VQ-DQN wins on parameter efficiency (34 vs 514 params).

Conclusions

1. **RLSTCcode achieves superior absolute clustering quality** (CR 0.59–0.82) due to 100× more training data and SGD with backpropagation gradients.
2. **Under identical SPSA training conditions**, the 34-parameter VQ-DQN outperforms all classical controls including one with 38× more parameters (1,314 → 1.64 CR vs 34 → 1.48 CR).
3. **The quantum advantage is optimizer-specific**: classical networks degrade catastrophically under SPSA (CR 5.11–9.21 for small/medium networks), while the quantum circuit structure provides sufficient inductive bias to learn effective policies.
4. **The data gap is the primary confound**: a fair same-data-same-optimizer comparison shows quantum winning. A fair same-data-same-optimizer-same-epochs comparison at scale remains an open experiment.

Plots

Unified Comparison

Figure 1: Unified Comparison

Data Scaling

Figure 2: Data Scaling

Controlled Comparison

Figure 3: Controlled Comparison

Parameter Efficiency

Figure 4: Parameter Efficiency

Summary Dashboard

Figure 5: Summary Dashboard